

A CLOSED-FORM MOMENT EXPANSION FOR A STIELTJES-TYPE MEASURE ASSOCIATED WITH $\zeta(\frac{1}{2} + it)$

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ABSTRACT. Let $\theta(t)$ be the Riemann–Siegel theta function and set $\Theta(t) := \theta(t) + Ct$ with $C > |\theta'(0)|$ so that $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing diffeomorphism. For a parameter $\alpha > 0$ define the density

$$h(t) := \zeta(\tfrac{1}{2} + it) \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\alpha} e^{i\Theta(t)},$$

and let Φ_G be the complex Borel measure on $\mathbb{R}$ obtained by pushing $h(t) dt$ forward along $u = \Theta(t)$. We give a closed form for every moment

$$\mu_n := \int_{\mathbb{R}} u^n d\Phi_G(u)$$

as an explicit finite sum in the Taylor coefficients $\zeta_k := \zeta^{(k)}(\frac{1}{2})$ and $\Theta_{2\ell+1} := \Theta^{(2\ell+1)}(0)$. The closed form is obtained by a single application of the residue / Lagrange inversion identity to the substitution $u = \Theta(t)$. We record the first seven moments in fully expanded form and discuss cumulants and moment-type consequences.

1. SETUP

Throughout, $\theta(t)$ is the Riemann–Siegel theta function,

$$\theta(t) = \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{\log \pi}{2} t,$$

and we fix a constant $C > |\theta'(0)| = |\frac{1}{2}\psi(\frac{1}{4}) - \frac{\log \pi}{2}|$ so that

$$\Theta(t) := \theta(t) + Ct$$

satisfies $\Theta'(t) > 0$ for all $t \in \mathbb{R}$. Because θ is odd and analytic at the origin, Θ is odd-plus-linear,

$$\Theta(t) = \sum_{\ell \geq 0} \frac{\Theta_{2\ell+1}}{(2\ell+1)!} t^{2\ell+1}, \quad \Theta_{2\ell+1} := \Theta^{(2\ell+1)}(0).$$

In particular $\Theta_1 = \frac{1}{2}\psi(\frac{1}{4}) - \frac{\log \pi}{2} + C > 0$. Let $\alpha > 0$ be a fixed convergence parameter.

Definition 1. The *G-measure* associated to ζ with parameters (C, α) is the pushforward under $u = \Theta(t)$ of the density

$$h(t) dt = \zeta(\tfrac{1}{2} + it) \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\alpha} e^{i\Theta(t)} dt.$$

Equivalently,

$$d\Phi_G(u) = \frac{h(t(u))}{\Theta'(t(u))} du,$$

where $t(u)$ is the (global real-analytic) inverse of Θ .

Since Θ is a diffeomorphism, $d\Phi_G$ is a well-defined complex measure on $\mathbb{R}$ as long as the $(1 + \Theta^2)^{-\alpha}$ weight renders the density integrable. The n th moment is

$$\mu_n := \int_{\mathbb{R}} u^n d\Phi_G(u) = \int_{\mathbb{R}} \Theta(t)^n h(t) dt.$$

2. THE CLOSED-FORM MOMENT IDENTITY

Theorem 2 (Closed form). *For every integer $n \geq 0$,*

$$(1) \quad \boxed{\mu_n \cdot \Theta_1^{n+\frac{1}{2}} = (-i)^n n! [t^n] \left\{ \zeta(\tfrac{1}{2} + it) \sqrt{\frac{\Theta'(t)}{\Theta_1}} e^{i\Theta(t)} (1 + \Theta(t)^2)^{-\alpha} \left(\frac{\Theta_1 t}{\Theta(t)} \right)^{n+1} \right\}},}$$

where $[t^n]$ denotes the coefficient of t^n in the formal power series obtained by expanding each factor at $t = 0$.

Proof. By definition

$$\mu_n = \int \Theta(t)^n h(t) dt = \int u^n d\Phi_G(u),$$

so

$$\mu_n = (-i)^n n! \cdot [u^n] \widehat{\Phi}_G(\xi) \Big|_{\xi \mapsto u},$$

where we identify μ_n with the coefficients in the formal moment-generating expansion $\sum_{n \geq 0} \mu_n (iu)^n / n! = \int e^{iu} d\Phi_G(u)$, viewed as a formal series in u .

Working on the level of formal power series near $u = 0$, write $F(u) := h(t(u))/\Theta'(t(u))$ so that $\mu_n = (-i)^n n! [u^n] F(u) \cdot (\text{const})$ up to the factor inherited from the Jacobian. The standard residue identity for a local diffeomorphism $u = \Theta(t)$ with $\Theta(0) = 0$, $\Theta'(0) \neq 0$, applied to any formal series $G(u)$, gives

$$[u^n] G(u) = \text{Res}_{u=0} \frac{G(u)}{u^{n+1}} du = \text{Res}_{t=0} \frac{G(\Theta(t)) \Theta'(t)}{\Theta(t)^{n+1}} dt = [t^n] G(\Theta(t)) \Theta'(t) \left(\frac{t}{\Theta(t)} \right)^{n+1}.$$

Applying this with $G(u) = e^{iu} (1 + u^2)^{-\alpha} \zeta(\frac{1}{2} + it(u)) / \sqrt{\Theta'(t(u))}$, one finds $G(\Theta(t)) = e^{i\Theta(t)} (1 + \Theta(t)^2)^{-\alpha} \zeta(\frac{1}{2} + it) / \sqrt{\Theta'(t)}$ and multiplying by $\Theta'(t)$ converts the $1/\sqrt{\Theta'}$ into $\sqrt{\Theta'}$. Scaling by $\Theta_1^{n+1/2}$ clears denominators and yields (??). $\square$

Remark 3. Formula (??) is a finite sum in any concrete use: each factor in the braces is a power series in t , each coefficient is a polynomial in α and in the Taylor coefficients ζ_k , $\Theta_{2\ell+1}$, and $[t^n]$ extracts a single finite combination. In particular, the only n -dependence in the formula is carried by the single binomial exponent $(n+1)$ in $(\Theta_1 t / \Theta(t))^{n+1}$.

3. EXPLICIT MOMENTS FOR $n \leq 7$

Let $\zeta_k := \zeta^{(k)}(\frac{1}{2})$. Expanding (??) through $n = 7$ yields:

$$\begin{aligned}
 \mu_1 \cdot \Theta_1^{3/2} &= \zeta_1 + \Theta_1 \zeta_0, \\
 \mu_2 \cdot \Theta_1^{5/2} &= \zeta_2 + 2\Theta_1 \zeta_1 + \left[(1 + 2\alpha)\Theta_1^2 + \frac{\Theta_3}{2\Theta_1}\right] \zeta_0, \\
 \mu_3 \cdot \Theta_1^{7/2} &= \zeta_3 + 3\Theta_1 \zeta_2 + \left[3(1 + 2\alpha)\Theta_1^2 + \frac{5\Theta_3}{2\Theta_1}\right] \zeta_1 + \left[(1 + 6\alpha)\Theta_1^3 + \frac{3\Theta_3}{2}\right] \zeta_0, \\
 \mu_4 \cdot \Theta_1^{9/2} &= \zeta_4 + 4\Theta_1 \zeta_3 + \left[6(1 + 2\alpha)\Theta_1^2 + \frac{7\Theta_3}{\Theta_1}\right] \zeta_2 + \left[4(1 + 6\alpha)\Theta_1^3 + 10\Theta_3\right] \zeta_1 \\
 &\quad + \left[(1 + 24\alpha + 12\alpha^2)\Theta_1^4 + 3(1 + 2\alpha)\Theta_1\Theta_3 + \frac{17\Theta_3^2}{4\Theta_1^2} - \frac{\Theta_5}{2\Theta_1}\right] \zeta_0, \\
 \mu_5 \cdot \Theta_1^{11/2} &= \zeta_5 + 5\Theta_1 \zeta_4 + \left[10(1 + 2\alpha)\Theta_1^2 + \frac{15\Theta_3}{\Theta_1}\right] \zeta_3 + \left[10(1 + 6\alpha)\Theta_1^3 + 35\Theta_3\right] \zeta_2 \\
 &\quad + \left[5(1 + 24\alpha + 12\alpha^2)\Theta_1^4 + 25(1 + 2\alpha)\Theta_1\Theta_3 + \frac{145\Theta_3^2}{4\Theta_1^2} - \frac{7\Theta_5}{2\Theta_1}\right] \zeta_1 \\
 &\quad + \left[(1 + 80\alpha + 60\alpha^2)\Theta_1^5 + 5(1 + 6\alpha)\Theta_1^2\Theta_3 + \frac{85\Theta_3^2}{4\Theta_1} - \frac{5\Theta_5}{2}\right] \zeta_0, \\
 \mu_6 \cdot \Theta_1^{13/2} &= \zeta_6 + 6\Theta_1 \zeta_5 + \left[15(1 + 2\alpha)\Theta_1^2 + \frac{55\Theta_3}{2\Theta_1}\right] \zeta_4 + \left[20(1 + 6\alpha)\Theta_1^3 + 90\Theta_3\right] \zeta_3 \\
 &\quad + \left[15(1 + 24\alpha + 12\alpha^2)\Theta_1^4 + 105(1 + 2\alpha)\Theta_1\Theta_3 + \frac{655\Theta_3^2}{4\Theta_1^2} - \frac{27\Theta_5}{2\Theta_1}\right] \zeta_2 \\
 &\quad + \left[6(1 + 80\alpha + 60\alpha^2)\Theta_1^5 + 50(1 + 6\alpha)\Theta_1^2\Theta_3 + \frac{435\Theta_3^2}{2\Theta_1} - 21\Theta_5\right] \zeta_1 \\
 &\quad + \left[(1 + 450\alpha + 540\alpha^2 + 120\alpha^3)\Theta_1^6 + \frac{15}{2}(1 + 24\alpha + 12\alpha^2)\Theta_1^3\Theta_3 - \frac{15}{2}(1 + 2\alpha)\Theta_1\Theta_5\right. \\
 &\quad \left.+ \frac{255(1+2\alpha)}{4}\Theta_3^2 + \frac{865\Theta_3^3}{8\Theta_1^3} - \frac{97\Theta_3\Theta_5}{4\Theta_1^2} + \frac{\Theta_7}{2\Theta_1}\right] \zeta_0,
 \end{aligned}$$

and

$$\begin{aligned}
 \mu_7 \cdot \Theta_1^{15/2} &= \zeta_7 + 7\Theta_1 \zeta_6 + \left[21(1 + 2\alpha)\Theta_1^2 + \frac{91\Theta_3}{2\Theta_1}\right] \zeta_5 + \left[35(1 + 6\alpha)\Theta_1^3 + \frac{385\Theta_3}{2}\right] \zeta_4 \\
 &\quad + \left[35(1 + 24\alpha + 12\alpha^2)\Theta_1^4 + 315(1 + 2\alpha)\Theta_1\Theta_3 + \frac{2135\Theta_3^2}{4\Theta_1^2} - \frac{77\Theta_5}{2\Theta_1}\right] \zeta_3 \\
 &\quad + \left[21(1 + 80\alpha + 60\alpha^2)\Theta_1^5 + 245(1 + 6\alpha)\Theta_1^2\Theta_3 + \frac{4585\Theta_3^2}{4\Theta_1} - \frac{189\Theta_5}{2}\right] \zeta_2 \\
 &\quad + \left[7(1 + 450\alpha + 540\alpha^2 + 120\alpha^3)\Theta_1^6 + \frac{175}{2}(1 + 24\alpha + 12\alpha^2)\Theta_1^3\Theta_3 - \frac{147}{2}(1 + 2\alpha)\Theta_1\Theta_5\right. \\
 &\quad \left.+ \frac{3045(1+2\alpha)}{4}\Theta_3^2 + \frac{10325\Theta_3^3}{8\Theta_1^3} - \frac{1015\Theta_3\Theta_5}{4\Theta_1^2} + \frac{9\Theta_7}{2\Theta_1}\right] \zeta_1 \\
 &\quad + \left[(1 + 2142\alpha + 2940\alpha^2 + 840\alpha^3)\Theta_1^7 + \frac{21}{2}(1 + 80\alpha + 60\alpha^2)\Theta_1^4\Theta_3 - \frac{35}{2}(1 + 6\alpha)\Theta_1^2\Theta_5\right. \\
 &\quad \left.+ \frac{595(1+6\alpha)}{4}\Theta_1\Theta_3^2 + \frac{6055\Theta_3^3}{8\Theta_1^2} - \frac{679\Theta_3\Theta_5}{4\Theta_1} + \frac{7\Theta_7}{2}\right] \zeta_0.
 \end{aligned}$$

All coefficients were verified by independent computation via Lagrange inversion of $u = \Theta(t)$ in symbolic form.

4. STRUCTURAL FEATURES

The ζ_{n-1} slot. The coefficient of ζ_{n-1} in $\mu_n \cdot \Theta_1^{n+1/2}$ is always $n\Theta_1$.

The leading α -polynomials. Setting $\Theta_3 = \Theta_5 = \Theta_7 = \dots = 0$ in (??) gives

$$\mu_n^{(0)} \cdot \Theta_1^{n+1/2} = n! \sum_{m=0}^n \frac{\zeta_m}{m!} \Theta_1^{n-m} \sum_{q=0}^{\lfloor (n-m)/2 \rfloor} \frac{(-1)^q \binom{-\alpha}{q}}{(n-m-2q)!}.$$

The innermost α -sum, call it $P_{n-m}(\alpha)$, produces the polynomials $P_0 = 1$, $P_1 = 1$, $P_2 = 1 + 2\alpha$, $P_3 = 1 + 6\alpha$, $P_4 = 1 + 24\alpha + 12\alpha^2$, $P_5 = 1 + 80\alpha + 60\alpha^2$, $P_6 = 1 + 450\alpha + 540\alpha^2 + 120\alpha^3$, $P_7 = 1 + 2142\alpha + 2940\alpha^2 + 840\alpha^3$.

Rational n -dependence. The only source of n -dependence in (??) is the exponent $n+1$ in $(\Theta_1 t / \Theta(t))^{n+1}$; hence every n -polynomial coefficient in the expanded moment is a sum of products of generalized binomials $\binom{-(n+1)}{Q}$ against universal rationals from the other four factors.

5. CONSEQUENCES

5.1. Cumulants. The logarithm of the moment-generating series of Φ_G provides the *cumulants* κ_n via $\log(1 + \sum_{n \geq 1} \mu_n z^n / n!) = \sum_{n \geq 1} \kappa_n z^n / n!$, so $\kappa_1 = \mu_1$, $\kappa_2 = \mu_2 - \mu_1^2$, $\kappa_3 = \mu_3 - 3\mu_2\mu_1 + 2\mu_1^3$, etc. Each κ_n inherits from Theorem ?? a closed finite-sum form in ζ_k and $\Theta_{2\ell+1}$.

5.2. Weighted second moment of $|\zeta|^2$. The pushforward of $|h(t)|^2 dt / \Theta'(t)$ under $u = \Theta$ is a genuine positive measure with moments $M_n := \int u^n |h(t(u))|^2 / \Theta'(t(u)) du$. Setting $n = 0$,

$$M_0 = \int_{-\infty}^{\infty} |\zeta(\tfrac{1}{2} + it)|^2 \Theta'(t) (1 + \Theta(t)^2)^{-2\alpha} dt,$$

which is a regularized version of the classical $\int_{-T}^T |\zeta(\tfrac{1}{2} + it)|^2 dt \sim T \log T$. The closed form of Theorem ??, applied to $Z\overline{Z}$ instead of Z , furnishes an analogous explicit expansion for every M_n .

5.3. Moment-generating function and zeros. The formal series $\sum_{n \geq 0} \mu_n z^n / n!$ is the Laplace transform of $d\Phi_G$. Its zeros in the complex plane, pushed back through Θ^{-1} on the real line, parameterize the zeros of $\zeta(\tfrac{1}{2} + it)$ weighted by $\sqrt{\Theta'(t)}(1 + \Theta^2)^{-\alpha} e^{i\Theta(t)}$. Theorem ?? gives every coefficient of this generating function in closed form.

5.4. A Carleman-type question. A moment sequence (μ_n) is determined by its measure if $\sum \mu_{2n}^{-1/(2n)} = \infty$ (Carleman's condition). Because every μ_n is now explicit in ζ_k and $\Theta_{2\ell+1}$, one can ask whether the growth rate of μ_n encodes RH. The closed form (??) gives $|\mu_n| \leq n! \cdot \Theta_1^{-n-1/2} \cdot C_n$ where C_n is a finite sum of absolute values of the listed rational-in- Θ_ℓ coefficients against $|\zeta_k|$. A precise Carleman / Hamburger analysis of these bounds, via the explicit formula above, is left for subsequent work.

6. REMARKS

The representation $d\Phi_G$ uses only the boundary values $\zeta(\tfrac{1}{2} + it)$, $t \in \mathbb{R}$. Formula (??) therefore gives, for each n , a finite linear combination of the Taylor data $\{\zeta^{(k)}(\tfrac{1}{2})\}_{k \leq n}$ which equals the integral $\int u^n d\Phi_G(u)$, an object that, on the face of it, depends only on the behavior of ζ on the critical line. This exchange of line-data Taylor coefficients for global integrals against the Θ -warped density is the key structural content.

Data availability. Symbolic verification of formula (??) and of every moment $\mu_1, \dots, \mu_7$ was performed with a SymPy computation; the short script reproduces the explicit table in Section 3 in a few seconds.